Closest Pair of Points – Brute Force

1. Aim

To implement the brute-force method in C to find the closest pair of points in a two-dimensional plane and analyze its time complexity.

2. Theory

The Closest Pair of Points problem involves finding two points whose Euclidean distance is the smallest among all possible pairs of points.

The brute-force approach compares every possible pair of points. For each pair, it calculates the Euclidean distance using the formula:

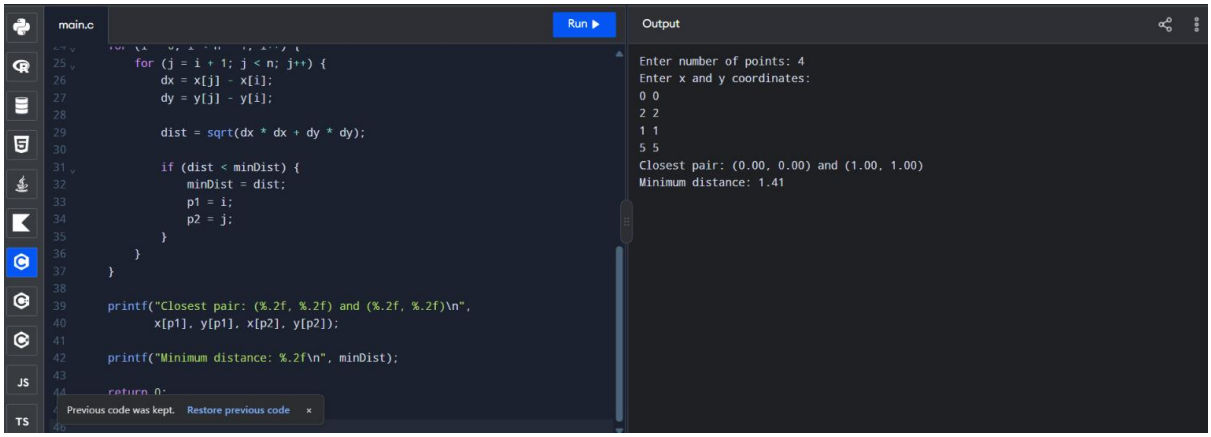
\[d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2} \]

The algorithm maintains the minimum distance found so far. Whenever a smaller distance is found, it updates the closest pair. After checking all pairs, it displays the two closest points and their distance.

This method is simple and guarantees the correct result, but it becomes slow when the number of points is large because it examines every pair.

3. Algorithm

1. Start.
2. Read the number of points and their coordinates.
3. Initialize the minimum distance to a very large value.
4. Select the first point.
5. Compare it with every point that follows it.
6. Calculate the Euclidean distance for each pair.
7. If the calculated distance is smaller than the current minimum, update the minimum distance and save the pair.
8. Repeat until every possible pair has been checked.
9. Display the closest pair and their minimum distance.
10. Stop.



6. Complexity Analysis

Let \(n\) be the number of points.

The algorithm checks all pairs of points. The total number of pairs is:

\[\frac{n(n-1)}{2} \]

- Best case: \(\Theta(n^2)\).
- Average case: \(\Theta(n^2)\).
- Worst case: \(\Theta(n^2)\).
- Space complexity: \(O(n)\) to store the coordinates; the additional working space is \(O(1)\).

The time complexity remains quadratic regardless of the arrangement of the points because the algorithm always examines every pair.

7. Result

The C program successfully identifies the closest pair of points using the brute-force approach and calculates their Euclidean distance.